

The Perceived Monetary Policy Rule: Identification and Macroeconomic Implications

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Motivation and Research Question

1. Taylor rule shows how monetary policy responds to macroeconomic conditions:

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Research Questions

1. *What are the macroeconomic implications of the perceived rule ($\neq$ actual rule)?*
2. *How do we identify the perceived rule $\phi_{t|t}$?*

Takeaways

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- **Equilibrium determinacy** depends on agents' *belief* in **Taylor principle**.
- Identifying the perceived rule requires interest rate **nowcasts**, not forecasts.
- The Taylor principle holds for **nowcast**- but not **forecast**-based estimates.
- **SP-MSNK model** separates the perceived rule from the actual rule.
 - *The Great Moderation*: managing perception matters.

A Simple Model: Why the Perceived Taylor Rule Matters

Setup:

$$i_t = E_t^P[\pi_{t+1}] + r_t, \quad r_t = \rho_r r_{t-1} + v_t$$

$$i_t = \phi_p \pi_t \quad (\text{perceived})$$

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How should we identify ϕ_p ?

Identification: Why Interest Rate Forecasts Introduce Bias

Monetary policy rule varies over time. (Kim-Nelson 2006, Coibion-Gorodnichenko 2011, Bianchi 2013):

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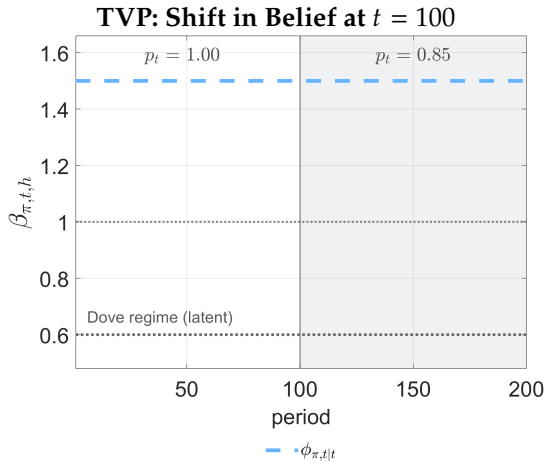
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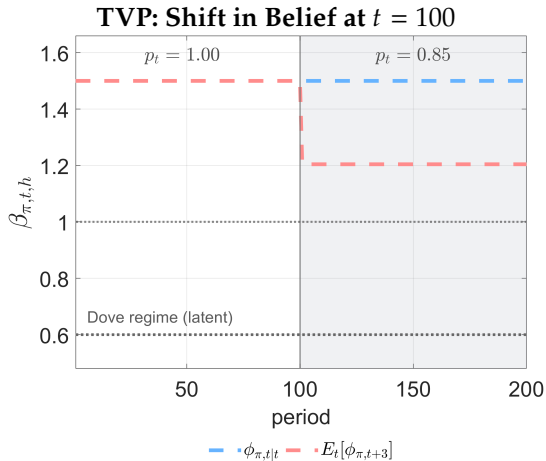
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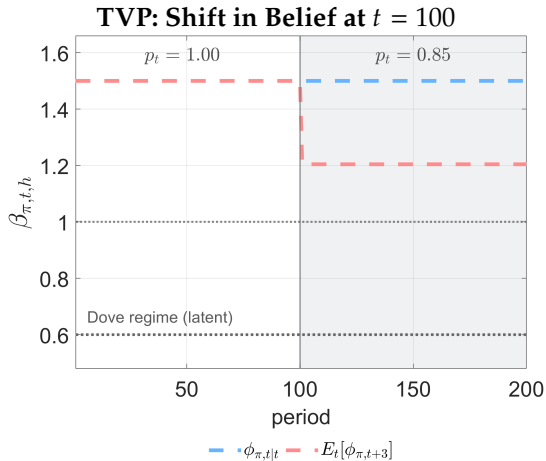
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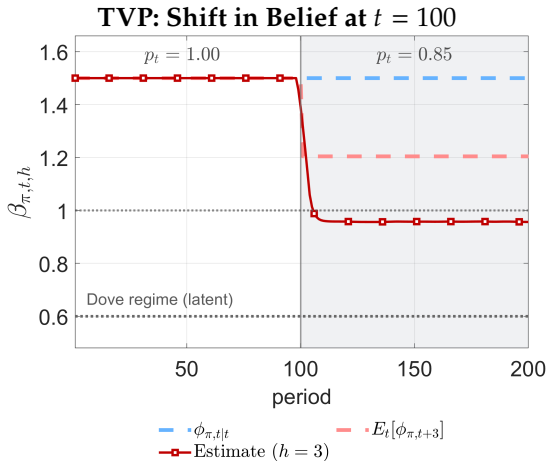
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$$\mathbb{E}_t[i_{t+h}] = \beta_{\pi,t,h} \mathbb{E}_t[\pi_{t+h}] + e_{t,h}$$

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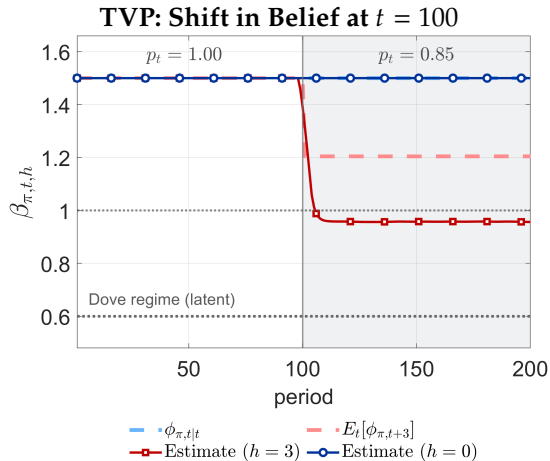
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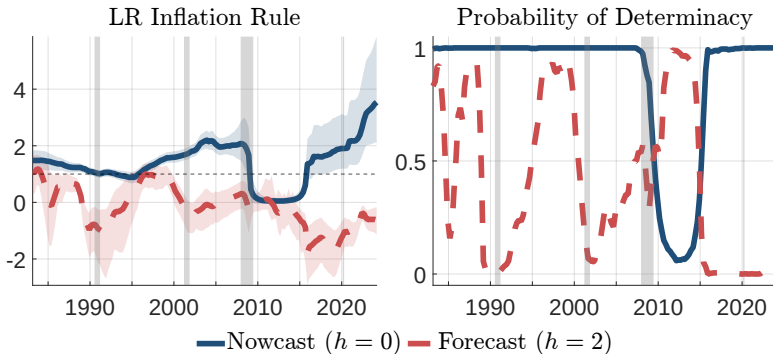
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Empirics: Nowcast-Based vs. Forecast-Based

Blue Chip + SPF survey expectations, 1983–2024 quarterly. TVP estimation of forward-looking rule with AR(2) inertia (Boivin 2005, Coibion-Gorodnichenko 2011).



- **Forecast**-based estimates are **attenuated** and **fail the Taylor principle**.
- **Nowcast**-based estimates are **stable** and **satisfy the Taylor principle**.

Note: Determinacy assessed under the textbook three-equation NK model.

SP-MSNK Model: Setup

Perceived model (Agent's subjective model):

$$x_t = E_t^P[x_{t+1}] - \tau(i_t - E_t^P[\pi_{t+1}]) + z_{Dt}$$

$$\pi_t = \beta E_t^P[\pi_{t+1}] + \kappa x_t + z_{St}$$

$$i_t = (1 - \rho_{s_t^p}^P) [\phi_{\pi, s_t^p}^P E_t^P[\pi_{t+1}] + \phi_{x, s_t^p}^P E_t^P[x_{t+1}]] + \rho_{1, s_t^p}^P i_{t-1} + \rho_{2, s_t^p}^P i_{t-2}$$

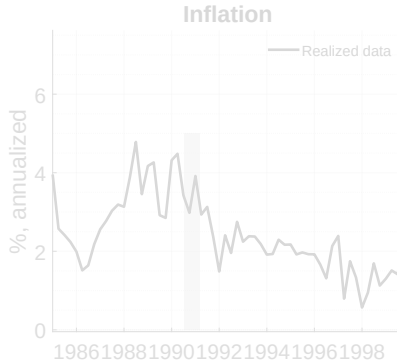
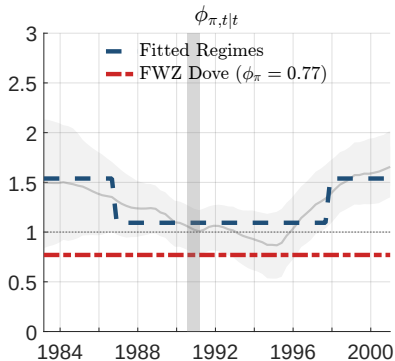
Actual Taylor rule (Fed's true reaction function):

$$i_t = (1 - \rho_{s_t^f}^F) [\phi_{\pi, s_t^f}^F E_t^P[\pi_{t+1}] + \phi_{x, s_t^f}^F E_t^P[x_{t+1}]] + \rho_{1, s_t^f}^F i_{t-1} + \rho_{2, s_t^f}^F i_{t-2} + \varepsilon_{Mt}$$

- Perceived regime s_t^p and the Fed's actual regime s_t^f .
- Subjective transitions $\hat{\mathbf{P}}$ of the perceived regime s_t^p .
- The implied ALM is TV-VAR:

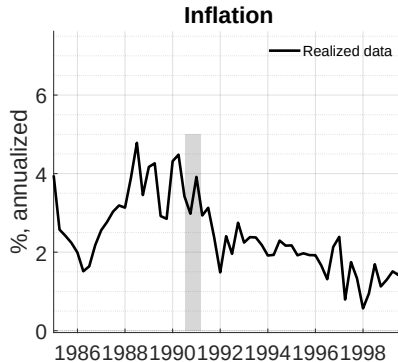
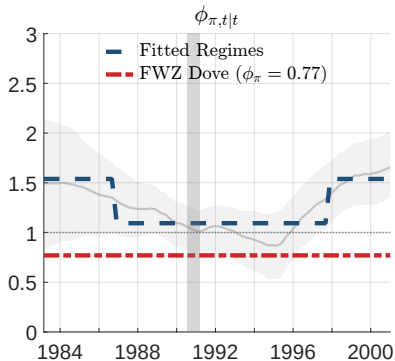
$$\mathbf{y}_t = \mathbf{A}_\ell^{\text{ALM}} \mathbf{y}_{t-1} + \mathbf{B}_\ell^{\text{ALM}} \varepsilon_t, \quad \ell = (s_t^p, s_t^f).$$

Counterfactual: Great Moderation



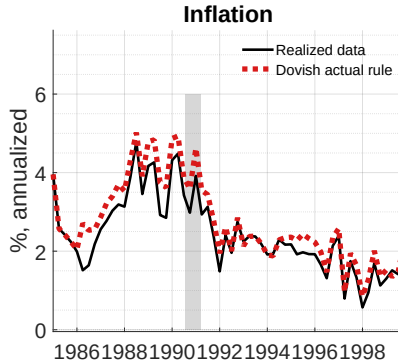
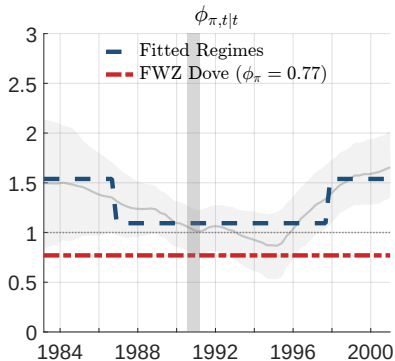
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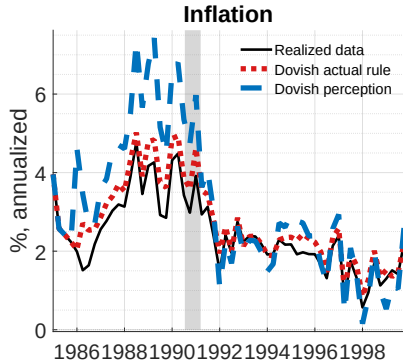
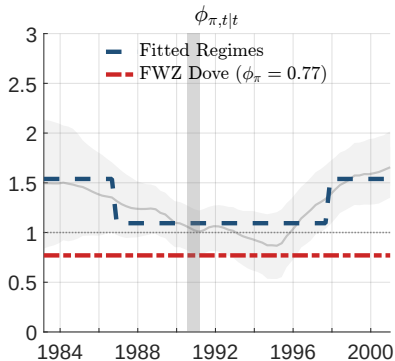
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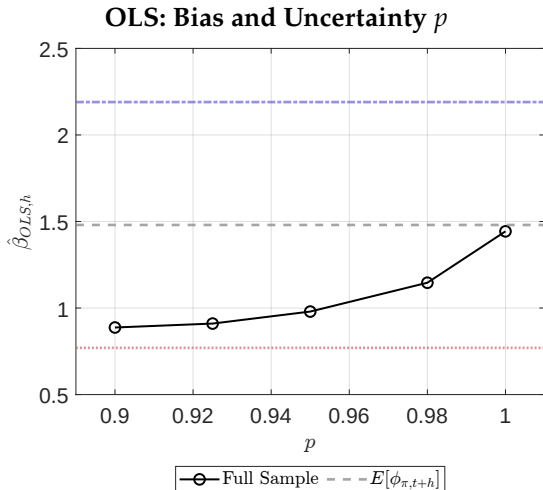
- *Dovish actual rule* (Hawkish perception) → modest rise.
- *Dovish perception* (Hawkish actual) → **large** volatility rise.

Conclusion. “**Good perceived policy**” underpinned the Great Moderation.

Conclusion

- The **perceived** monetary policy rule governs **macroeconomic dynamics**.
 - Determinacy depends on the perceived Taylor principle.
 - The Great Moderation required a shift in perception, not just the actual rule.
- **Measuring** perception requires interest rate **nowcasts**, not forecasts.
- **Managing** perception is key to effective policy.

Simulation (Backup): OLS Bias and Regime Uncertainty p



- Unconditional OLS on $\mathbb{E}_t[i_{t+h}] = \beta_h \mathbb{E}_t[\pi_{t+h}] + e_{t,h}$.
- Downward bias increases in regime uncertainty p .
- Spurious Taylor-principle violation at the forecast horizon.
- What matters is the agent's uncertainty, not a realised regime shift.